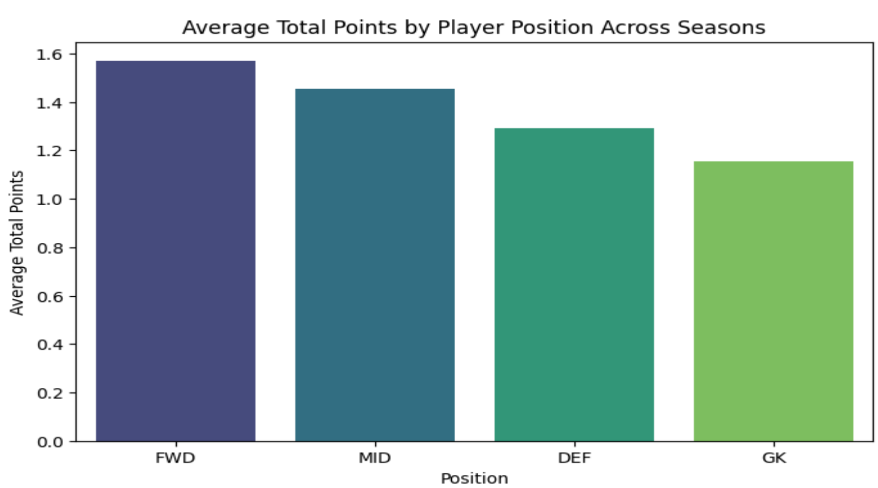
Milestone 1 Analytical Report – Fantasy Premier League (FPL)

1. Data Engineering Analysis

1.1 Average Total Points by Player Position

Position	Average Total Points
FWD	1.57
MID	1.45
DEF	1.29
GK	1.16

Interpretation.
Forwards produced the highest average total points, followed closely by midfielders. Defenders and goalkeepers contributed less on average. This distribution reflects how FPL scoring prioritizes attacking contributions (goals, assists, and bonus points) that are far more common among attacking players. A bar chart of these averages clearly shows the steep drop from forward to defensive roles, highlighting the positional imbalance in FPL point potential.



1.2 Top Five Players by Total Points (2022–23 Season)

1. Erling Haaland
2. Harry Kane
3. Mohamed Salah
4. Martin Ødegaard
5. Marcus Rashford

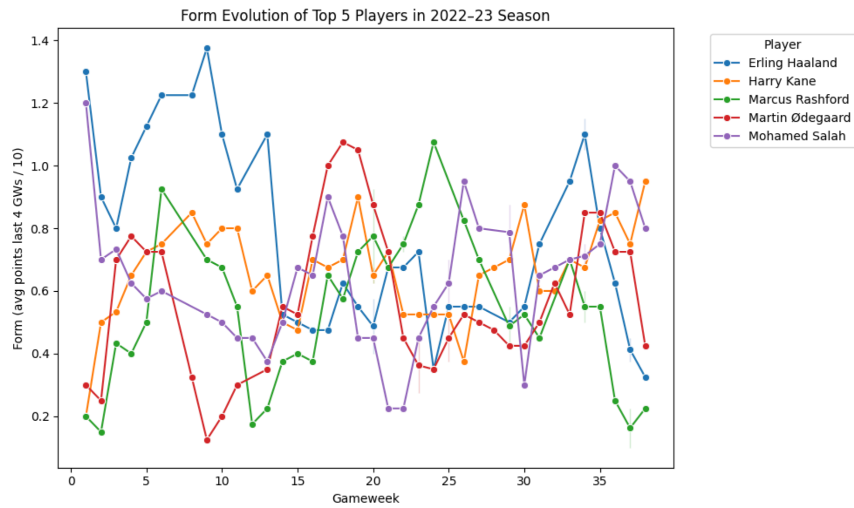
Interpretation.
The list is dominated by elite forwards and midfielders from high-scoring clubs. Haaland and Kane lead because of their consistent goal-scoring and 90-minute appearances. The visualization of total points confirms how much these two strikers outperformed the rest of the player pool.

1.3 Top Five Players by Average Form

Player	Average Form
Erling Haaland	0.75
Harry Kane	0.66
Mohamed Salah	0.64
Martin Ødegaard	0.56
Gabriel Martinelli Silva	0.55

Interpretation.
The top-form ranking largely mirrors the total-points list, showing that players who consistently maintain high form metrics also accumulate the most season-long points. Plots of form evolution across gameweeks show sharp rises for Haaland early in the season and steady performance

from Kane and Salah, indicating reliability rather than volatility.



1.4 Overall Data Insights

Visual analyses of position averages and top-player trends demonstrate a clear pattern: attacking output drives FPL performance. Goal involvement and playing time are the strongest quantitative indicators of fantasy success, guiding both feature selection and model design in the predictive stage.

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2. Predictive Model Features and Rationale

The model was trained to predict **`upcoming_total_points`** using the following features:

Feature	Rationale
`total_points`	Historical performance is a baseline indicator of a player's potential for future points.
`minutes`	Playing time directly limits opportunities to earn any points. Consistent starters are more predictable.
`goals_scored`	One of the primary FPL scoring components and a strong predictor of future returns.
`assists`	Captures playmaking contribution that complements goal scoring.

`bonus` and `bps`	Represent a player's likelihood of receiving FPL bonus points based on in-match performance metrics.
`clean_sheets`	Important for defenders and goalkeepers; contributes additional points when teams concede few goals.
`creativity`, `influence`, `threat`, `ict_index`	Underlying FPL metrics summarizing attacking involvement and contribution beyond goals/assists. They quantify how "active" a player is in producing chances.
`form`	Recent short-term performance trend; adds recency awareness to the model.
`value`	Price efficiency; often correlates with both quality and manager selection rate.
`was_home`	Home advantage can significantly increase scoring likelihood.
`yellow_cards`, `red_cards`	Penalties and suspensions reduce expected future returns.
`position`	Contextualizes the scoring system (e.g., defenders gain from clean sheets, forwards from goals).
`team_x`	Team strength and playing style influence all other performance indicators.

Together, these variables capture a balanced view of productivity, opportunity, and risk for each player, combining objective statistics and contextual information.

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3. Predictive Model and Performance Summary

Model Type

A **Linear Regression** model was trained to predict `upcoming_total_points`.Data was split into training and testing subsets (80 / 20). This baseline regression was chosen for its interpretability and ability to measure linear contribution of each feature to future FPL returns.

Results

Metric	Value

MAE	1.3666
MSE	5.2413
RMSE	2.2894
R^2	0.3073

Interpretation.
The R^2 value of 0.31 indicates that roughly one-third of the variance in upcoming total points can be explained by the chosen features. The mean absolute error of 1.37 suggests the model's predictions are generally within one to two FPL points of actual outcomes, which is reasonable for a dataset containing significant weekly variability.

Observations

- The positive relationship between attacking metrics (goals, assists, threat) and predicted points confirms their predictive power.
- Defensive metrics (clean sheets, cards) have weaker, often negative, contributions for attackers, showing role-specific differences.
- Despite simplicity, the linear model establishes a clear quantitative link between recent performance and future returns, providing a transparent baseline before experimenting with non-linear or ensemble approaches.

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4. Conclusion

This milestone's data engineering and predictive modeling steps reveal how feature engineering and exploratory analysis guide performance prediction in Fantasy Premier League data. Attacking roles dominate the scoring structure, and recent form combined with cumulative statistics produces the most reliable predictors of future points.